RTTI - One way streets

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

RTTI - One way streets

Description

This conceptual framework describes the information needed to record where motorised traffic may travel in one direction only. The core is the permitted direction of travel: on a one-way road, travel is allowed in a single direction only. That direction is fixed relative to the road itself: forward or return. Exceptions often apply. A very common one is ‘except cyclists’, where cyclists may ride in both directions. There are also exceptions for, for example, scheduled buses or access-only traffic. Sometimes the one-way restriction applies only at certain times, for instance during rush hour or at events; this is called a dynamic one-way road. Sometimes the direction of travel even changes. A reversible lane may be open in the forward direction at the start of the day and in the return direction at the end of the day. For each one-way situation, the framework records: where the rule applies (on which road section) and in which direction; how road users perceive the rule: traffic signs along the road, supplementary plates showing exceptions or times, and advance warning signs; when the rule applies: always or only at certain times, and, where the direction of travel alternates, which direction applies when; which road users are exempt; what the direction of travel is based on and, for an established direction of travel, which traffic decree applies. The framework covers every situation in which motorised traffic may travel in one direction only. For each situation, the basis of the direction of travel is recorded. In many cases, a road authority deliberately establishes the one-way restriction by traffic decree and makes the restriction known through traffic signs; exemptions and time periods may apply. On other roads, the direction of travel follows from the traffic rules and the road layout, as on the separate carriageways of motorways and other dual carriageways, on slip roads and on roundabouts. Such an inherent direction of travel has no traffic decree and no exemptions. These situations nevertheless fall within the framework, because driving against the direction of travel can have serious consequences there, and the signage intended to prevent entry, such as ‘no entry’ signs at exit ramps, matters for road safety. A reversible lane always has an established direction of travel: there, the direction of travel does not follow from the traffic rules but is set by the road authority. One-way traffic is recorded because the permitted direction of travel determines which movements across the road network are possible and safe. The data support route planning per transport mode: a car or freight route follows the one-way restriction, whereas a cycle route makes use of an exemption for cyclists. The data contribute to road safety by enabling in-vehicle systems and traffic control centres to detect driving against the direction of travel and to issue warnings. For reversible lanes and dynamic one-way restrictions, the data show which direction applies at which moment. The link to the traffic decree and the signage makes it possible to verify whether the situation on the road matches the decree. Because all of this data is described in the same way, road authorities, navigation systems and traveller information services can exchange and use the data reliably. The framework was drawn up within the European agreements on travel and traffic information (ITS Directive 2010/40/EU and RTTI Regulation (EU) 2022/670); the RTTI Regulation lists one-way streets and the direction of travel on reversible lanes as traffic regulations to be made available. The framework aligns with existing Dutch sources for roads and traffic signs: NDW and the Nationaal Wegenbestand (NWB, the Dutch national road database).

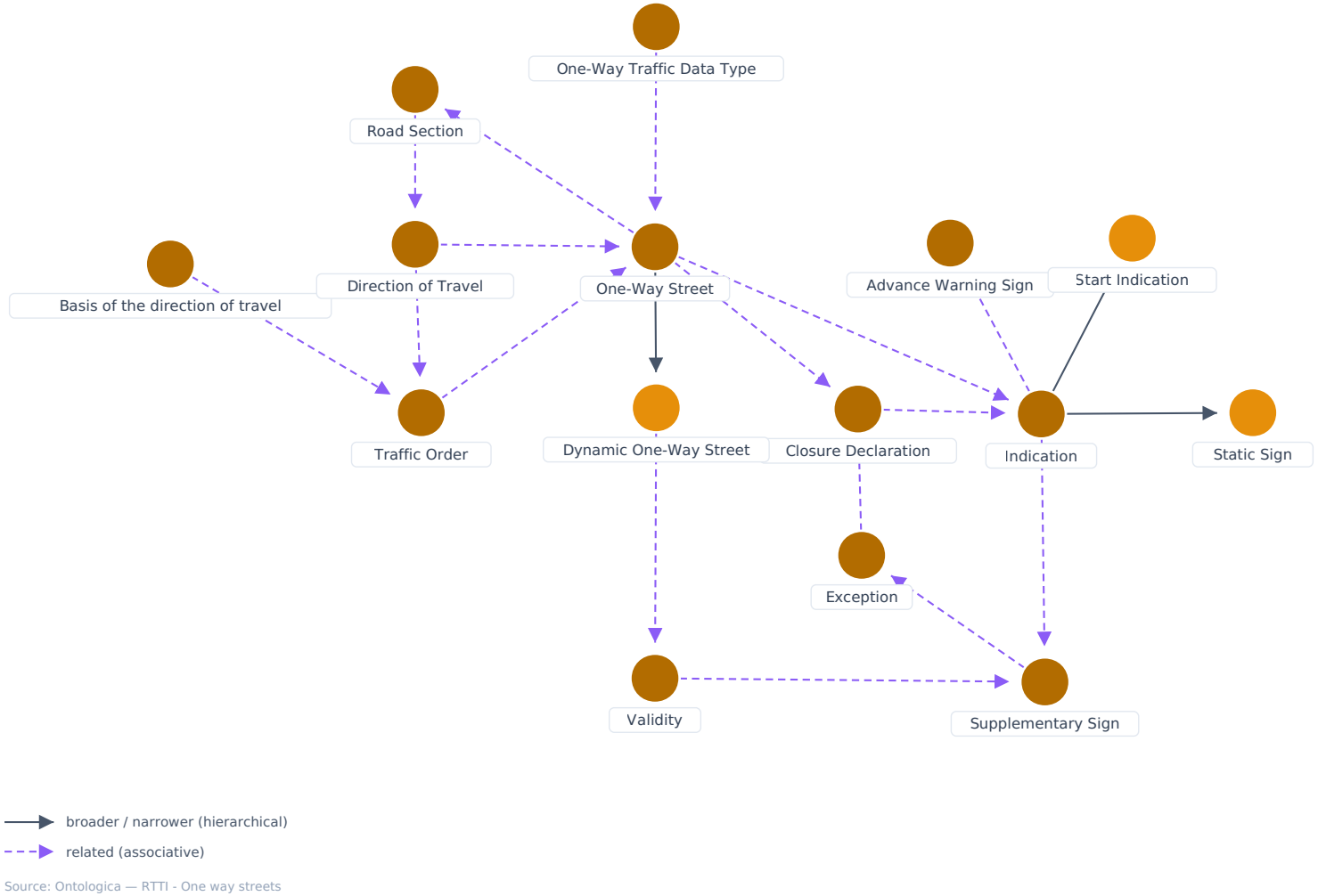
Metadata

Field	Value
URI	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064

Field	Value
Version	0.7.0
Publication status	Under review (Under development)
Management status	Active
Organization	Nationaal Toegangspunt Mobiliteitsdata
Created on	2026-02-27
Last modified on	2026-10-29
Last modified by	arthur.keesen@ndw.nu

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Source: Ontologica — RTTI - One way streets

Concepts

Basis of the direction of travel

Concept details

Field	Value
Definition	Characteristic indicating what the permitted direction of travel of a one-way situation is based on: a traffic decree of the competent authority, or the traffic rules in combination with the road layout.
Scope	Determines which other data apply. An established direction of travel requires a traffic decree and any exemptions. Where the direction of travel derives from the traffic rules and road layout, no traffic decree exists and no exemptions apply; the direction of travel is then taken from the Nationaal Wegenbestand (NWB) and the closure declaration intended to prevent entry against the direction of travel is recorded.. A reversible lane always has an established direction of travel. Not to be confused with the legal basis, which is the traffic decree itself.
Example	A one-way street in a residential area, established by a municipal traffic decree: established by traffic decree. The right-hand carriageway of the A2 motorway: derived from traffic rules and road layout.

One-Way Traffic Data Type

Concept details

Field	Value
Definition	Data type within the RTTI context describing the set of concepts required to unambiguously record and exchange one-way traffic regulations on the Dutch road network. Encompasses the direction of travel, the basis of the direction of travel, the indication, any exemptions and, where applicable, the traffic decree.

One-Way Street

Concept details

Field	Value
Definition	A road segment where traffic is allowed to travel in only one direction.
Scope	A one-way street rests on one of two bases (see Basis of the direction of travel). An established one-way street is set by a traffic decree and indicated by a combination of traffic signs. The core is sign C2 (one-way street, closed in this direction) on the closed side and sign C3 (one-way street) on the open side. In addition, sign C4 and mandatory direction signs (chapter D of Annex 1 to the RVV 1990), among others, can support the regulation. Supplementary signs indicate exemptions or times. The traffic signs used are described under Indication. For an inherent direction of travel, as on dual carriageways, slip roads and roundabouts, no traffic decree exists and no exemptions apply; a closure declaration intended to prevent entry against the direction of travel may be present. The one-way regulation is recorded as the direction-of-travel attribute on one or more consecutive NWB road sections; start and end follow from the boundaries of those sections.

Road Section

Concept details

Field	Value
Definition	A section of road between two consecutive junctions or intersections, or – if no side road is present – between two points to which a traffic measure applies. In the National Road Database (NWB) a road section is the smallest unit for which road attributes are registered.

Direction of Travel

Concept details

Field	Value
Definition	The direction or directions in which traffic is permitted to travel along a section of road.
Scope	Within this vocabulary, the direction of travel is the characteristic that distinguishes a one-way street from a two-way street. In the National Road Database (NWB), the direction of travel is recorded per road section, relative to the digitisation direction of that section. The attribute is essential for routing on the NWB and is also used in the NDW WKD files to record restrictions per direction. The NWB attribute does not distinguish the basis of the direction of travel: road sections with an established direction of travel and road sections with an inherent direction of travel, such as slip roads, dual carriageways and roundabouts, are coded as one-way in the same way. Both fall within this framework; the distinction is recorded with Basis of the direction of travel. Exceptions per road user category, such as ‘except cyclists’, are not included in the direction-of-travel attribute. See Exception.

Closure Declaration

Concept details

Field	Value
Definition	A closure declaration prohibiting traffic from entering a road segment in a specific direction. Indicated by sign C2 (RVV 1990): one-way street, closed in this direction for vehicles, riders and handlers of draught or pack animals or livestock.
Scope	Sign C2 is the primary indication of a one-way street. The closure declaration applies to all vehicles, riders and handlers, unless an exemption is indicated by a supplementary sign. In practice, ‘except cyclists and moped riders’ is the most common exemption. Sign C2 also occurs with an inherent direction of travel, for example at the end of an exit ramp, to prevent entry against the direction of travel.

Indication

Concept details

Field	Value
Definition	The traffic sign or combination of signs by which a one-way traffic regulation is communicated to road users.

Static Sign

Concept details

Field	Value
Definition	A physical ('tin') traffic sign permanently placed to indicate the one-way traffic regulation.

Start Indication

Concept details

Field	Value
Definition	The traffic sign indicating the point at which the one-way regulation takes effect.

Supplementary Sign

Concept details

Field	Value
Definition	A sign mounted below a traffic sign to provide additional specification of the scope of that traffic sign.

Advance Warning Sign

Concept details

Field	Value
Definition	An advance warning sign placed upstream of the one-way street, at a junction where road users have the opportunity to choose an alternative route.

Exception

Concept details

Field	Value
Definition	A condition that modifies the standard one-way regulation, allowing certain road user categories to travel in the otherwise closed direction. Exceptions are communicated via supplementary signs.

Validity

Concept details

Field	Value
Definition	The time period during which the one-way regulation applies.

Dynamic One-Way Street

Concept details

Field	Value
Definition	A road segment where the one-way regulation is temporary or changeable, imposed in response to time of day, events, or prevailing conditions.

Traffic Order

Concept details

Field	Value
Definition	A formal traffic order constituting the legal basis for imposing a one-way traffic regulation.
Scope	Traffic regulations for one-way streets currently consist largely of unstructured text. While these are published via KOOP/LVBB, a structured link to specific traffic signs and locations is not yet available as a standard feature. Only applicable in the context of basis of the direction of travel.

Relations per concept

Outgoing semantic relations only — these stay inside the model.

Basis of the direction of travel

Semantic relations

Type	Target label	Target concept
related	Traffic Order	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000015

One-Way Traffic Data Type

Semantic relations

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002

One-Way Street

Semantic relations

Type	Target label	Target concept
narrower	Dynamic One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000014
related	Closure Declaration	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000005
related	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006
related	Road Section	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000003

Road Section

Semantic relations

Type	Target label	Target concept
related	Direction of Travel	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000004

Direction of Travel

Semantic relations

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002
related	Traffic Order	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000015

Closure Declaration

Semantic relations

Type	Target label	Target concept
related	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Indication

Semantic relations

Type	Target label	Target concept
narrower	Static Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000007

Type	Target label	Target concept
narrower	Start Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000008
related	Advance Warning Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000011
related	Supplementary Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000010

Static Sign

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Start Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Supplementary Sign

Semantic relations

Type	Target label	Target concept
related	Exception	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000012

Exception

Semantic relations

Type	Target label	Target concept
related	Closure Declaration	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000005

Validity

Semantic relations

Type	Target label	Target concept
related	Supplementary Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000010

Dynamic One-Way Street

Semantic relations

Type	Target label	Target concept
broader	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002
related	Validity	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000013

Traffic Order

Semantic relations

Type	Target label	Target concept
related	Direction of Travel	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000004

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002
related	Basis of the direction of travel	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-1793262693650